

Error Analysis — the “menstrual” interpretation gap

Question raised by the baseline report: the base model interprets menstrual concerns at only ~0.39 while fertility is ~0.90 (§4.3). Is this a model blind spot, or a labeling artifact? **Answer: predominantly a category-boundary artifact — the “errors” are largely clinically-defensible reads of inherently overlapping cases.**

All figures from `M2m1_{en,fr,ar}.jsonl` scored by `scripts/evaluate.py`.

1. What menstrual-gold items get predicted as

Confusion for the 180 menstrual-gold items per language (base model M2, `parse_ok`):

Predicted	EN	FR	AR
menstrual (correct)	0.39	0.39	0.40
fertility	0.38	0.36	0.36
PCOS	0.14	0.20	0.20
other	0.08	0.04	0.04

The misreads are not random: **87.2%** of the 109 wrong EN menstrual items are mapped to an **adjacent category (fertility or PCOS)**; only **12.8%** go to “other”. The model isn’t failing to understand — it is disambiguating an overlapping case toward a different, usually reasonable, category.

2. Why: the menstrual seeds are conception- and PCOS-themed

14 of 30 menstrual seeds (47%) explicitly mention conception/pregnancy. These are labeled `menstrual` (presenting symptom = irregular periods) but their *chief complaint* / *patient intent* is fertility. Actual inputs the model “got wrong”:

Seed	Gold	Model	Input (verbatim)
menst-002	menstrual	fertility	“Do irregular periods influence the ability to get pregnant?”
menst-008	menstrual	fertility	“Due to my irregular periods I am worried about getting pregnant ... Is it possible for me to conceive even with irregular periods?”
menst-018	menstrual	fertility	“married for 4yrs, but i havnt conceived, my menstrual cycle is sometimes painful ... desperate need a child”
menst-019	menstrual	fertility	“missed my periods by 3 days and the test on the strip came weakly positive ... could I be pregnant?”
menst-021	menstrual	fertility	“35 year old married woman ... trying for child since 1 yr ... i missed periods ...”

Seed	Gold	Model	Input (verbatim)
menst-023	menstrual	fertility	“want to be pregnant because of my irregular periods my dr suggest me to take siphene ...”
menst-030	menstrual	PCOS	“irregular periods. After an ultrasound ... I have multiple cysts over the periphery of ovaries. I want to conceive ...”

- The **fertility** confusions are conception-motivated period complaints — the model keys on the stated goal (“want to be pregnant”, “trying to conceive”) rather than the presenting symptom. Clinically this is a defensible, arguably more useful, read.
- The **PCOS** confusions (e.g. menst-030) are textbook PCOS presentations — *irregular periods + polycystic ovaries on ultrasound* — where the label **menstrual** is debatable and **PCOS** is arguably **more correct** than the gold.

12 menstrual seeds are wrong on **all 6 phrasings** in every language — i.e. it is a stable property of the *case/label*, not of surface form or language.

3. Cross-model: fine-tuning shifts the confusion, doesn’t fix it

Model	Lang	menstrual interp	dominant wrong-prediction
M2 (base)	EN/FR/AR	0.39 / 0.39 / 0.40	fertility (69/65/65)
M3ml-v1	FR/AR	0.37 / 0.38	PCOS (83/80)
M3ml-v2	EN/FR/AR	0.37 / 0.33 / 0.38	PCOS (84/82/74)

Fine-tuning **moves the menstrual confusion from fertility (base) toward PCOS**, because the adaptation corpus reshapes the model’s category priors — but the underlying overlap is not resolved by any model. This is itself a finding: the low menstrual score is a property of the **taxonomy/labels interacting with real patient language**, robust across models and languages.

4. Implications for the benchmark and paper

1. **Raw category accuracy understates true interpretation quality.** ~87% of menstrual “errors” are clinically-adjacent, and a non-trivial share (the conception- and PCOS-themed seeds) are reasonable or arguably correct.
2. **Real patient messages do not respect clean category boundaries.** Half the menstrual seeds are conception-motivated; irregular-periods-with-ovarian-cysts is simultaneously “menstrual” and “PCOS”. This is a genuine language-understanding phenomenon, not just noise.
3. **Recommended actions for the paper (pick one or combine):**
 - Report a **relaxed / clinically-acceptable interpretation** metric that credits adjacent-category reads (menstrual↔fertility when conception is the stated intent; menstrual↔PCOS when polycystic ovaries are described), alongside strict accuracy.
 - **Refine the gold labels** for the ~14 conception-motivated menstrual seeds (multi-label, or relabel by chief complaint), then re-score.
 - Keep strict accuracy but **explicitly frame the menstrual gap as category-boundary ambiguity** with this confusion analysis as evidence — turning a weak-looking number into a substantive finding.
4. **Do not** claim the base model “cannot understand menstrual concerns” — the evidence contradicts it. The correct claim is that menstrual presentations in real patient language frequently carry fertility or PCOS intent that the model (reasonably) surfaces.

5. Reproduce

*# confusion + examples were produced with scripts/evaluate.py scorers over
Used_Datasets/Consolidated_Datasets/200_Seed_Dataset/M2ml_{en,fr,ar}.jsonl
(gold_category vs predicted_category on gold_category == "menstrual").*